THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 3 & 4: **The World of Metals and Non-metals, Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** ductility	e.g. copper wire
2	**(b)** Graphite	form of carbon
3	**(a)** good conductors of heat and electricity	typical property
4	**(b)** painting or galvanising	keeps air and water away
5	**(c)** kerosene	reacts with air and water
6	**(d)** physical change	water remains water
7	**(b)** Cooking an egg	new substances
8	**(a)** combustible	e.g. wood, LPG
9	**(b)** chemical change	colour, taste change
10	**(c)** physical change	returns to shape

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It reacts vigorously with air and water (2).	2
12	Metals are good conductors of heat (2).	2
13	A change that can be undone (1); freezing of water (1).	1+1
14	A change that cannot be reversed (1); burning of wood (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Mixtures of metals/non-metals (1.5); to improve strength and resistance to corrosion (1.5).	1½+1½
16	Physical; original metal returns (3).	3
17	Dissolve in water (1); metal oxide turns red litmus blue (1); non-metal oxide turns blue litmus red (1).	1+1+1
	OR: Keeps moisture and air away to prevent rust (3).	3
18	Burning gunpowder forms new gases and heat (3).	3
	OR: Curd, idli, bread (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Mixture (1). (ii) Steel, brass, bronze (2). (iii) Stronger, resist corrosion (2).	1+2+2
	OR: (i) Nail, wire, lead (1). (ii) Straw (1). (iii) Graphite is a conducting non-metal (1). (iv) Electrical wiring (2).	1+1+1+2
20	(i) Formation of crystals from solution (1). (ii) Evaporation (2). (iii) Yes (2).	1+2+2
	OR: (i) Both chemical, but one has no heat (1). (ii) Bioluminescence (1). (iii) Very little (1). (iv) Light without heat (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

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SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** Sulfur	a yellow non-metal
2	**(a)** oxygen	life support
3	**(d)** sodium	stored in kerosene
4	**(a)** corrosion	gradual deterioration
5	**(c)** stainless steel	alloy
6	**(b)** carbon dioxide	turns lime water milky
7	**(a)** Bursting of a firecracker	explosion
8	**(d)** a chemical change	combustion
9	**(c)** Combustion	burning
10	**(b)** a chemical change	new substance

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Good conductors and ductile (2).	2
12	Oxygen and carbon (2).	2
13	Desirable: ripening of fruit; undesirable: food spoilage (2).	2
14	Carbon dioxide reacts to give calcium carbonate (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	It is brittle and a poor conductor (3).	3
16	Rusting is slow without flame; burning is fast with flame (3).	3
17	Metals are sonorous (3).	3
	OR: Iron—tools, copper—wires, aluminium—foil; carbon—fuel, sulfur—matches, oxygen—breathing (3).	3
18	Useful but harms the environment (3).	3
	OR: Oil floats and spreads, making the fire worse (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Definitions (2). (ii) Aluminium, silver, gold; copper, aluminium (2). (iii) Gold (1).	2+2+1
	OR: (i) Basic and acidic (2). (ii) Litmus test (2). (iii) MgO; SO_2 (1).	2+2+1
20	(i) Based on time taken (2). (ii) Slow: weathering, rusting; fast: lightning, firecrackers (2). (iii) Breaks rocks (1).	2+2+1
	OR: (i) Gives heat and energy (1). (ii) Pollution and global warming (2). (iii) Use clean fuels, save energy (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET C

Chapters 3 & 4: **The World of Metals and Non-metals, Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** sonority	e.g. bells
2	**(a)** diamond	a form of carbon
3	**(d)** poor conductors of heat and electricity	except graphite
4	**(c)** corrosion	reaction with moist air
5	**(d)** a white powder	magnesium oxide
6	**(b)** chemical change	new substances formed
7	**(a)** magnesium oxide	white powder
8	**(a)** bioluminescence	fireflies
9	**(d)** chemical change	breakdown into simpler substances
10	**(d)** evaporation	physical process

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Burns with a bright white flame forming white magnesium oxide (2).	2
12	A mixture of a metal with other elements (1); steel (1).	1+1
13	Slow: weathering, soil formation; fast: lightning, burning (2).	2
14	Fizzing as carbon dioxide gas is formed (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Damages bridges, ships and vehicles (1.5); costs money to repair (1.5).	1½+1½
16	Physical (1); crystallisation from solution (2).	1+2
17	Keep in kerosene (1); handle with forceps (1); keep away from water (1).	1+1+1
	OR: Does not rust (1.5); strong and shiny (1.5).	1½+1½
18	Both are chemical changes producing new substances (3).	3
	OR: Only state differs (1.5); composition is the same (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Keep air and water away (1). (ii) Painting, oiling, galvanising (3). (iii) Saves money and resources (1).	1+3+1
	OR: (i) Salt helps corrosion (1). (ii) Paint (1). (iii) Zinc blocks (1). (iv) Sacrificial protection/galvanisation (2).	1+1+1+2
20	(i) Spoilage, chemical (1). (ii) No (1). (iii) Refrigeration (1). (iv) No (2).	1+1+1+2
	OR: (i) Can or cannot be undone (2). (ii) Yes by evaporation (1.5). (iii) No (1.5).	2+1½+1½

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET D

Chapters 3 & 4: **The World of Metals and Non-metals, Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** Aluminium	a metal
2	**(a)** iodine	antiseptic
3	**(d)** sonorous	ring when struck
4	**(d)** copper	metals conduct
5	**(a)** Aluminium and iron	check periodic classification
6	**(a)** carbon dioxide	test for CO_2
7	**(b)** Food getting spoiled	harmful
8	**(a)** iron oxide	reddish-brown
9	**(c)** chemical	new substance
10	**(d)** a reversible physical change	salt is recovered

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Graphite (1); in pencil leads and electrodes (1).	1+1
12	Metal oxides basic (1); non-metal oxides acidic (1).	1+1
13	Only shape changes; no new substance forms (2).	2
14	Cuts off oxygen supply or cools the fuel (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Metals conduct heat; wood and plastic do not (3).	3
16	Fuels with a low value catch fire easily (1.5); kerosene more easily than paper (1.5).	1½+1½
17	Iron conducts; plastic is an insulator (3).	3
	OR: Prevents rusting (3).	3
18	Digestion; respiration (3).	3
	OR: Cold light through chemical reactions (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Heat passes along the metal (2). (ii) Rod with wax and pins (2). (iii) Plastic is a poor conductor (1).	2+2+1
	OR: (i) Table with observations (3). (ii) Classification (2).	3+2
20	(i) New substance (1). (ii) Magnesium + oxygen → magnesium oxide + heat + light (2). (iii) Magnesium oxide (2).	1+2+2
	OR: (i) Desirable: ripening, curd, cooking; undesirable: spoilage, rusting, pollution (3). (ii) Prevention (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.